Huawei Access Node WHI-ACC-50 Whitefield East POP Hardware/Environmental Fault Resolution SOP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7058

Description

The Huawei Access Node WHI-ACC-50 at the Whitefield East POP is prone to hardware/environmental faults due to the presence of ASIC memory parity errors and line card crashes. This is primarily attributed to the intrinsic nature of the Huawei's Access Node architecture, where the line cards are directly integrated with the ASICs, thereby creating a single point of failure. Furthermore, the loss of power supply redundancy can exacerbate this issue, causing the Access Node to malfunction. The ALARM_7058 alarm code, indicative of a minor severity, signals the occurrence of these hardware/environmental faults. It is crucial to promptly diagnose and rectify these faults to prevent any potential service disruptions. In this SOP, we outline the steps to be taken to identify and resolve the root cause of the hardware/environmental faults in the Huawei Access Node WHI-ACC-50 at the Whitefield East POP.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield East POP to facilitate physical access to the equipment.

- Verify the status of the redundant links to ensure they are operational.

- Check the power supply redundancy to confirm it is functioning correctly.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-manage fault-record, Expected Output: Fault ID | Fault Type | Fault Location | Fault Severity | Fault Time, Description: This command is used to retrieve the fault records from the Access Node. It helps in identifying the type, location, and severity of the faults.

Command: display device power-supply, Expected Output: Power Supply ID | Current Status | Power Supply Type | Voltage, Description: This command is used to retrieve the power supply information from the Access Node. It helps in verifying the power supply redundancy.
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```
Command: display board info, Expected Output: Board ID | Board Type | Board Status | Temperature, Description: This command is used to retrieve the board information from the Access Node. It helps in identifying any issues with the line cards.
```

Resolution Steps

Step 1: Step: Replace the Faulty Line Card, Command: replace board [Board ID], Description: This step involves replacing the faulty line card with a new one. The replace board command is used to replace the faulty line card with a new one.

Step 2: Step: Restore Power Supply Redundancy, Command: redundant power-supply enable, Description: This step involves restoring the power supply redundancy. The redundant power-supply enable command is used to restore the power supply redundancy.

Step 3: Step: Clear Fault Records, Command: clear fault-manage fault-record, Description: This step involves clearing the fault records from the Access Node. The clear fault-manage fault-record command is used to clear the fault records.

Post-Check Verification

```
Command: display fault-manage fault-record, Expected Output: No fault records found, Description: This command is used to verify if the fault records have been cleared.

Command: display device power-supply, Expected Output: Power Supply ID | Current Status | Power Supply Type | Voltage, Description: This command is used to verify if the power supply redundancy has been restored.

Command: display board info, Expected Output: Board ID | Board Type | Board Status | Temperature, Description: This command is used to verify if the line card has been replaced.
```

Roll-back Procedures

- Step: Rollback to Previous Configuration, Command: undo replace board [Board ID], Description: This step involves rolling back to the previous configuration. The undo replace board command is used to roll back to the previous configuration.

- Step: Disable Power Supply Redundancy, Command: redundant power-supply disable, Description: This step involves disabling the power supply redundancy. The redundant power-supply disable command is used to disable the power supply redundancy.

- Step: Restore Fault Records, Command: restore fault-manage fault-record, Description: This step involves restoring the fault records from the Access Node. The restore fault-manage fault-record command is used to restore the fault records.